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Financial Analytics Lab

GROUP 11

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Answers

[Link to EXCEL SHEET](#)

Q1. Answer to each part of question 1 is as follows

NPV = INR 819027253.75

IRR= 16%

Payback Period = 4 Years

Profitability Index = 1.09

Yes, Sigma should take the contract and open the mine because of a positive NPV and a higher IRR than the required rate of return.

All metrics are in favour of taking the contract

Cashflows in Each Year									
Time	Cost	Production	Revenues	Benefits	WC	CF (before tax)	Cumulative Profit	CF (after tax)	PV
0	₹ 8,46,90,00,000.00				₹ 23,17,20,000.00	-₹ 8,70,07,20,000.00	-₹ 8,46,90,00,000.00	-₹ 8,70,07,20,000.00	-₹ 8,70,07,20,000.0
1	₹ 2,29,31,20,000.00	₹ 6,20,000.00	₹ 4,63,44,00,000.00		₹ 25,23,30,000.00	₹ 2,34,12,80,000.00	-₹ 6,12,77,20,000.00	₹ 2,32,06,70,000.00	₹ 2,07,20,26,785.7
2	₹ 2,47,76,80,000.00	₹ 6,80,000.00	₹ 5,04,66,00,000.00		₹ 26,95,05,000.00	₹ 2,56,89,20,000.00	-₹ 3,55,88,00,000.00	₹ 2,55,17,45,000.00	₹ 2,03,42,35,491.0
3	₹ 2,63,14,80,000.00	₹ 7,30,000.00	₹ 5,39,01,00,000.00		₹ 46,05,75,000.00	₹ 2,75,86,20,000.00	-₹ 80,01,80,000.00	₹ 2,56,75,50,000.00	₹ 1,82,75,31,375.2
4	₹ 2,20,08,40,000.00	₹ 5,90,000.00	₹ 9,21,15,00,000.00			₹ 7,01,06,60,000.00	₹ 6,21,04,80,000.00	₹ 5,42,17,76,600.00	₹ 3,44,56,37,046.3
5	₹ 26,70,00,000.00					-₹ 26,70,00,000.00		-₹ 26,70,00,000.00	-₹ 15,15,02,970.48
6				₹ 57,60,00,000.00		₹ 57,60,00,000.00		₹ 57,60,00,000.00	₹ 29,18,19,525.80
								NPV	₹ 81,90,27,253.75
								IRR	16%
								Payback Period (In Years)	4
								Profitbaility Index	1.09

Answers

Q2. All the revenues and costs are discounted to 1 Jan 2002.

A). **Price of water** = Rs. 0.1954723 / m3

B). **i. NPV** = Rs. -379202.0938

ii. IRR = 5.244474 %

	Interest Rate	NPV
	0.0524448	-0.9267445852
	0.0524447	0.397163274
IRR (approx)	**0.05244474**	**-0.000009366078302**

Q3. Tender B should be preferred over Tender A if the maximum NPV criterion is followed.

The discounting rates and time periods were adjusted for semi-annual payments, that is r → r/2 and t → 2t, and then discounting was done.

The completion time for both tenders was different, so the benefit of 80 million was also discounted to present value accordingly.

Answers

Time (in months)	Tender A (Amount in millions)	Present Value of Tender A cash outlfows	Tender B (Amount in millions)	Present Value of Tender B cash outflows
0	5	5	2.5	2.5
6	5	4.734176017	10	9.468352033
12	5	4.645683386	15	13.44745353
18	5	4.244174134	5	4.244174134
24	5	4.018533479	5	4.018533479
30	5	3.804888964		
36	5	3.602602816	5	3.602602816
42	5	3.411071169		
48	5	3.229722264	7.5	4.844583396
54	5	3.058014737		
60				
	PV of all Cash outflows for tender A	39.74886696	PV of all Cash outflows for tender B	42.12569939
Net Benefits of the project (in millions)	80	46.32697608	80	51.67555623
	NPV for tender A	**6.578109116**	**NPV for tender B**	**9.549856836**
	Since NPV for tender B > tender A, Tender B must be preferred			